

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

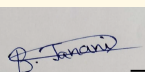
ASSESSMENT TOOL 1 – COMPARITIVE ANALYSIS

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO4	Assessment:	AT1 - COMPARITIVE ANALYSIS
Weightage:	30 % of CIA	Total Marks:	100
CO4 Statement:	Identify the components and structure of mobile application for Android and Windows OS-based devices.		
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Section / Batch:	Slot A- 2023	Date:	18.08.2026

Code of Conduct

I B Janani 192321100 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____



Instructions

- Duration: 60 minutes
- Number of questions: 01

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A - QUESTIONS

1. A software company wants to develop a mobile application that can run across devices manufactured by different vendors. Compare Android and Windows mobile platforms in terms of architecture, hardware abstraction, development frameworks, and application deployment. Analyze which platform is more suitable for multi-vendor device support.

Comparative Analysis of Android and Windows Mobile

Introduction

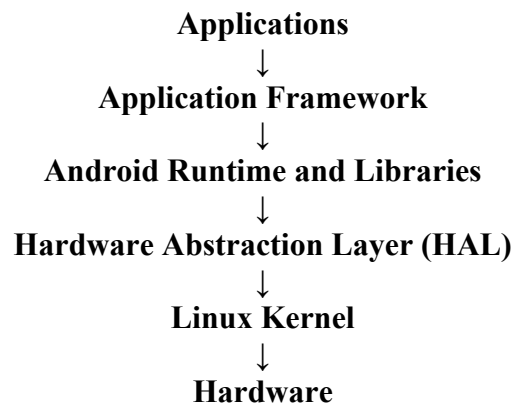
A software company wants to develop a mobile application that can work on devices made by different manufacturers. For this purpose, the operating system should support different hardware, provide easy development tools, and allow simple application deployment.

Android and Windows Mobile can be compared based on their architecture, hardware abstraction, development frameworks, and application deployment.

1. Architecture

Android

Android is mainly based on the **Linux kernel**. Its architecture is divided into different layers:



The Linux kernel handles important functions such as:

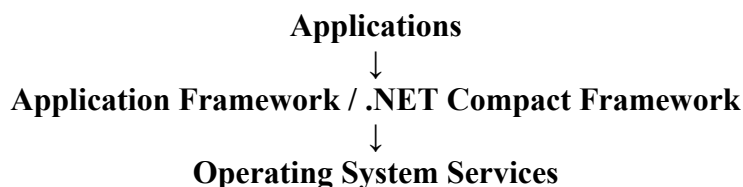
- Memory management
- Process management
- Security
- Device drivers
- Networking
- Power management

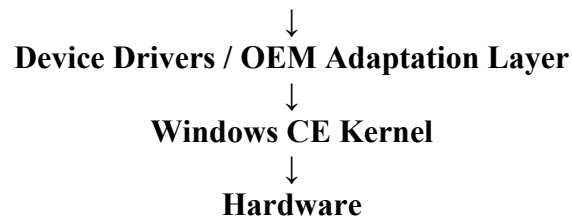
Android uses a layered architecture, so applications do not need to directly communicate with the hardware.

Windows Mobile

Traditional Windows Mobile is mainly based on the Windows CE architecture.

Its simplified architecture is:





The Windows CE kernel manages memory, processes, devices, networking, and other system functions.

Comparison

Android has a more flexible architecture because manufacturers can adapt it to different hardware configurations. Windows Mobile also supports different devices, but it depends more on Microsoft and manufacturer-specific implementations.

2. Hardware Abstraction

Android

Android uses a Hardware Abstraction Layer (HAL).

HAL acts as a bridge between the Android operating system and the device hardware.

For example, different mobile manufacturers may use different:

- Cameras
- GPS modules
- Bluetooth chips
- Sensors
- Processors

The application does not need to know the exact hardware used in each phone. It communicates through standard Android APIs.

The flow is:

Application → Android API → HAL → Device Driver → Hardware

This makes Android suitable for devices produced by many different manufacturers.

Windows Mobile

Windows Mobile uses device drivers and an OEM Adaptation Layer to communicate with hardware.

Manufacturers must provide suitable drivers and components for their devices.

It provides hardware abstraction, but the level of flexibility is lower compared with Android.

3. Development Frameworks

Android

Android provides a large and flexible development environment.

Main tools and technologies include:

- Android Studio
- Android SDK
- Java
- Kotlin
- C/C++ using Android NDK

Android provides APIs for accessing:

- Camera
- GPS
- Bluetooth
- Wi-Fi
- Sensors

- Storage
- Multimedia
- Notifications

Developers can use the same Android APIs on devices from different manufacturers.

Windows Mobile

Windows Mobile applications were mainly developed using:

- Microsoft Visual Studio
- .NET Compact Framework
- C#
- Visual Basic
- C/C++
- Windows Mobile SDK

It provided good support for developers who were already familiar with Microsoft technologies.

However, its development environment and device ecosystem were more limited compared with Android.

4. Application Deployment

Android

Android applications can be distributed in several ways.

They can be installed through:

- Google Play Store
- Manufacturer app stores
- Third-party app stores
- Enterprise distribution
- Direct installation or sideloading

Android applications are commonly distributed using APK or Android App Bundle (AAB) formats.

This provides developers with more freedom when distributing applications.

Windows Mobile

Windows Mobile applications could be distributed using:

- Microsoft-supported distribution methods
- Enterprise deployment
- Direct installation
- CAB packages in traditional Windows Mobile systems

Windows Mobile provided useful enterprise deployment options, but its application ecosystem was smaller and less flexible.

5. Overall Comparison

Feature	Android	Windows Mobile
Kernel	Linux Kernel	Windows CE Kernel
Hardware abstraction	Hardware Abstraction Layer	OEM Adaptation Layer / Drivers
Hardware flexibility	High	Moderate
Main languages	Kotlin, Java, C/C++	C#, VB.NET, C/C++
Development tool	Android Studio	Visual Studio
Development framework	Android SDK	.NET Compact Framework
Device	Supports many	Supports fewer

manufacturers	vendors	vendors
Application distribution	Highly flexible	More limited
Customization	High	Moderate
Multi-vendor support	Excellent	Limited compared with Android

6. Which Platform is Better for Multi-Vendor Devices?

For this software company, Android is more suitable for multi-vendor device support.

The main reason is Android's Hardware Abstraction Layer.

Different manufacturers may use different hardware, but Android applications can still use common Android APIs.

For example, one manufacturer may use one type of camera while another manufacturer uses a different camera. The application can still use the same Android camera API because HAL and device drivers handle the hardware differences.

Android also provides:

- Support for many device manufacturers
- Flexible hardware integration
- Standard APIs
- Large development framework
- Easy application distribution
- Greater customization

One disadvantage of Android is device fragmentation. Devices may have different screen sizes, hardware, Android versions, and manufacturer modifications. Therefore, developers must test the application on different devices.

Even with this limitation, Android provides much better support for different hardware manufacturers.

Conclusion

Both Android and Windows Mobile provide mobile application development frameworks and hardware abstraction mechanisms.

Windows Mobile provides a structured Microsoft-based environment and good support for enterprise applications. However, it has comparatively limited hardware and manufacturer support.

Android provides a Linux-based architecture, Hardware Abstraction Layer, standard APIs, flexible development tools, and multiple application deployment methods.

Therefore, Android is the better choice for developing an application that must run across devices manufactured by different vendors.

Final Result

Android is more suitable for multi-vendor device support because it provides better hardware flexibility, standard APIs, wider manufacturer support, and easier application deployment.